

Propositional Logic Patterns

Reworked to match the source sheet layout: multiple columns and rows, with conclusions beneath premises. Greek metavariables replaced by P, Q, R , biconditional kept as $\leftrightarrow$.

Disjunctive Syll. 1 (DS1) $P \vee Q$ $\neg P$ $\therefore Q$	Disjunctive Syll. 2 (DS2) $P \vee Q$ $\neg Q$ $\therefore P$	Disjunctive Syll. 3 (DS3) $\neg P \vee Q$ P $\therefore Q$	Disjunctive Syll. 4 (DS4) $P \vee \neg Q$ Q $\therefore P$
Modus Tollens 1 (MT1) $P \rightarrow Q$ $\neg Q$ $\therefore \neg P$	Modus Tollens 2 (MT2) $P \rightarrow \neg Q$ Q $\therefore \neg P$	Modus Tollens 3 (MT3) $\neg P \rightarrow Q$ $\neg Q$ $\therefore P$	Modus Tollens 4 (MT4) $\neg P \rightarrow \neg Q$ Q $\therefore P$
Exclusion 1 (EX1) $\neg(P \wedge Q)$ P $\therefore \neg Q$	Exclusion 2 (EX2) $\neg(P \wedge Q)$ Q $\therefore \neg P$	Exclusion 3 (EX3) $\neg(P \wedge \neg Q)$ P $\therefore Q$	Exclusion 4 (EX4) $\neg(\neg P \wedge Q)$ Q $\therefore P$
DeMorgan 1 (DM1) $\neg(P \vee Q)$ $\therefore \neg P \wedge \neg Q$	DeMorgan 2 (DM2) $\neg P \wedge \neg Q$ $\therefore \neg(P \vee Q)$	DeMorgan 3 (DM3) $\neg(P \wedge Q)$ $\therefore \neg P \vee \neg Q$	DeMorgan 4 (DM4) $\neg P \vee \neg Q$ $\therefore \neg(P \wedge Q)$
Half DeMorgan 1 (HDM1) $\neg(P \vee Q)$ $\therefore \neg P$	Half DeMorgan 2 (HDM2) $\neg(P \vee Q)$ $\therefore \neg Q$	Double Negation (DN) $\neg\neg P$ $\therefore P$	Law of Excluded Middle (LawEM) $\therefore P \vee \neg P$
Hypothetical Syllogism (HS) $P \rightarrow Q$ $Q \rightarrow R$ $\therefore P \rightarrow R$	Resolution (Resolution) $P \vee Q$ $\neg Q \vee R$ $\therefore P \vee R$	Constructive Dilemma (CD) $P \vee Q$ $P \rightarrow R$ $Q \rightarrow S$ $\therefore R \vee S$	Destructive Dilemma (DD) $\neg R \vee \neg S$ $P \rightarrow R$ $Q \rightarrow S$ $\therefore \neg P \vee \neg Q$

Strengthening the Antecedent 1 (Strength1) $P \rightarrow R$ $\therefore (P \wedge Q) \rightarrow R$	Strengthening the Antecedent 2 (Strength2) $P \rightarrow R$ $\therefore (Q \wedge P) \rightarrow R$	Strengthening the Antecedent 3 (Strength3) $(P \vee Q) \rightarrow R$ $\therefore P \rightarrow R$	Strengthening the Antecedent 4 (Strength4) $(P \vee Q) \rightarrow R$ $\therefore Q \rightarrow R$
Weakening the Consequent 1 (Weak1) $P \rightarrow (Q \wedge R)$ $\therefore P \rightarrow Q$	Weakening the Consequent 2 (Weak2) $P \rightarrow (Q \wedge R)$ $\therefore P \rightarrow R$	Weakening the Consequent 3 (Weak3) $P \rightarrow Q$ $\therefore P \rightarrow (Q \vee R)$	Weakening the Consequent 4 (Weak4) $P \rightarrow Q$ $\therefore P \rightarrow (R \vee Q)$
Negation Conditional Elimination 1 (NCE1) $\neg(P \rightarrow Q)$ $\therefore P$	Negation Conditional Elimination 2 (NCE2) $\neg(P \rightarrow Q)$ $\therefore \neg Q$	Negation Conditional Introduction 1 (NCI1) $\neg P$ $\therefore P \rightarrow Q$	Negation Conditional Introduction 2 (NCI2) Q $\therefore P \rightarrow Q$
BiConditional Elimination 1 (BE1) $P \leftrightarrow Q$ $\neg P$ $\therefore \neg Q$	BiConditional Elimination 2 (BE2) $P \leftrightarrow Q$ $\neg Q$ $\therefore \neg P$	Negation BiConditional Elimination 1 (NBE1) $\neg(P \leftrightarrow Q)$ P $\therefore \neg Q$	Negation BiConditional Elimination 2 (NBE2) $\neg(P \leftrightarrow Q)$ Q $\therefore \neg P$
BiConditional Introduction 1 (BI1) P Q $\therefore P \leftrightarrow Q$	BiConditional Introduction 2 (BI2) $\neg P$ $\neg Q$ $\therefore P \leftrightarrow Q$	Negation BiConditional Introduction 1 (NBI1) $\neg P$ Q $\therefore \neg(P \leftrightarrow Q)$	Negation BiConditional Introduction 2 (NBI2) P $\neg Q$ $\therefore \neg(P \leftrightarrow Q)$
Negation BiConditional 1 (NB1) $\neg(P \leftrightarrow Q)$ $\therefore P \leftrightarrow \neg Q$	Negation BiConditional 2 (NB2) $\neg(P \leftrightarrow Q)$ $\therefore \neg P \leftrightarrow Q$	Negation BiConditional 3 (NB3) $P \leftrightarrow \neg Q$ $\therefore \neg(P \leftrightarrow Q)$	Negation BiConditional 4 (NB4) $\neg P \leftrightarrow Q$ $\therefore \neg(P \leftrightarrow Q)$

BiConditional Commutation 1 (BC1) $P \leftrightarrow Q$ $\therefore Q \leftrightarrow P$	BiConditional Commutation 2 (BC2) $\neg(P \leftrightarrow Q)$ $\therefore \neg(Q \leftrightarrow P)$	BiConditional Converse 1 (BCon1) $P \leftrightarrow Q$ $\therefore \neg P \leftrightarrow \neg Q$	BiConditional Converse 2 (BCon2) $\neg P \leftrightarrow \neg Q$ $\therefore P \leftrightarrow Q$
BiConditional Converse 3 (BCon3) $P \leftrightarrow \neg Q$ $\therefore Q \leftrightarrow \neg P$	BiConditional Converse 4 (BCon4) $\neg P \leftrightarrow Q$ $\therefore \neg Q \leftrightarrow P$	BiConditional Converse 5 (BCon5) $P \leftrightarrow \neg Q$ $\therefore \neg P \leftrightarrow Q$	BiConditional Converse 6 (BCon6) $\neg P \leftrightarrow Q$ $\therefore P \leftrightarrow \neg Q$